

# RAJA KUMAR

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## SUMMARY

Final year B.Tech CSE student (CGPA 9.07) specializing in AI/ML and intelligent systems. Experienced in building RAG-based applications and scalable solutions with strong foundations in DSA and system design.

## SKILLS

**Programming Languages:** C, Java, Python

**Frameworks & Libraries:** Next.js, React, Express, TensorFlow, NumPy, Tailwind CSS, Asternity

**Tools & Platforms:** Git, VS Code, PyCharm, IntelliJ IDEA, Jupyter Notebook

**AI/ML and Data Science:** Machine Learning, Deep Learning, NLP, Generative AI, LLMs

**Core Competencies:** Data Structures, Algorithms, OOP, DBMS, Software Engineering, Project Management

## PROJECTS

**MIRAI: Autonomous AI Combat System** | Python, FastAPI, FAISS, Ollama, Next.js GitHub

- Engineered a memory-augmented AI combat system inspired by NVIDIA ACE using FAISS vector retrieval and behavioral profiling.
- Integrated N-gram prediction, heuristic strategy engines, and emotional state modeling for adaptive combat behavior.
- Designed a resource-aware combat engine with stamina, posture, parry, and stagger mechanics for dynamic AI interactions.
- Connected Ollama with Llama 3.2 and architected a Next.js dashboard interface for adaptive dialogue and combat telemetry visualization.

**Quant Agent: Multi-Agent Trading System** | Python, LangGraph, LangChain, Flask GitHub

- Developed a multi-agent AI trading platform using LangGraph for automated market analysis and decision orchestration.
- Integrated technical indicators including RSI, MACD, and Bollinger Bands, improving trend prediction accuracy by 30%.
- Utilized LLMs such as GPT-4o, Claude, and Qwen for context-aware trade analysis, reducing false signals by 25%.
- Engineered scalable Flask APIs for real-time data processing, model inference, and inter-agent communication.

**PickItUp: Hyperlocal Shop Reservation Platform** | Next.js, React.js, MongoDB GitHub (In Progress)

- Created a hyperlocal reservation platform with GPS-based shop discovery using Geolocation API and MongoDB geospatial queries.
- Implemented scalable modules for inventory management, merchant dashboards, reservations, and OTP-based pickup verification.
- Optimized MongoDB aggregation pipelines with MongoDB 2dsphere indexing for efficient real-time radius search and high-performance API responses.

## EDUCATION

**KIIT UNIVERSITY** 2023 - 2027

B.Tech in Computer Science and Engineering

**CGPA: 9.07**

Relevant Coursework: Object-Oriented Programming, Databases, Discrete Mathematics, Data Structures, Algorithms, Operating Systems, Computer Networks, Machine Learning, Data Mining

## OTHER EXPERIENCE

**SwasthyaPro** Jul 2025 – Sep 2025

*Digital Marketing Intern (Remote)*

- Improved organic search visibility and analyzed 1000+ monthly traffic insights using Google Analytics and SEMrush to optimize SEO performance.

## ACHIEVEMENTS & CERTIFICATIONS

- Completed AWS Cloud Architecting and Data Science, AI/ML & Generative AI Specialization.
- Solved 200+ DSA problems and secured 1st Runner-Up at KIIT Hackathon for building the UiPath Resume Screener.